

Abhijeet Vikram

BS-MS Interdisciplinary Sciences

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Education

- **Bachelor and Master of Science (BSMS) in Interdisciplinary Sciences** - 5th year (expected 2027)
Indian Institute of Science Education and Research, Pune, India
 - **The Aryan International School, Varanasi, India**
CBSE Class XII (2022): 93% | Class X (2020): 92%
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Relevant Coursework

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| <ul style="list-style-type: none">• Linear Algebra• Probability• Calculus I & II• Statistical Inference• Markov Chain and its Application• Inverse Theory• Non-Linear Dynamics• Graph Theory• Stochastic Processes (auditing) | <ul style="list-style-type: none">• Introduction to Computing• Statistical Learning and Data Science• Deep Learning• Image & Video Processing with Deep Learning• NLP• Signal Processing |
| <ul style="list-style-type: none">• Mathematical Finance• Economics and Public Policy | <ul style="list-style-type: none">• Neurobiology I & II• Bioinformatics• Systems Biology• Ecology• Chemical ecology• Animal Behaviour• Genome Biology |
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Research Experience

- **Master's Thesis (May-Apr 2027): A study on Low-Precision Quantization and Pruning of State Space Models for Edge Computing applications like Brain-Computer Interface (Prof. Chetan Singh Thakur - IISc Bangalore)**

Working on multiple fronts in brain signal decoding, mainly focusing on building very small models (<10,000 parameters) for NeuroAI tasks like seizure prediction and hand motion prediction with sEEG and ECoG data.

Working on ECoG to audio models to convert noisy brain ECoG data into hearable natural language audio and be deployable on edge devices. Both works are still in progress and under review.

- **Semester Project (Aug-Apr 2025): Diffusion model with deterministic Noising (Prof. MS Santhanam - IISER Pune) - [Report](#)**

To study diffusion models and make them more energy and compute efficient **by replacing the stochastic noising process with deterministic chaotic dynamics, making the reversal process faster** and better suited for edge devices in compute constrained environments.

The reversal process still remains a difficult problem due to the loss of information and the symmetric nature of the logistic map, in contrast to the noise accumulation seen in conventional diffusion models. This attempt has still provided deep insights into nonlinear dynamics and generative AI. Still working on it.

- **Summer Project (May-July 2025): Spiking Harmonic REsonate and fire SSM Model (SHaRe - SSM) (Prof. Ayon Borthakur - IIT Guwahati)**
 - We proposed and rigorously evaluated a **second-order spiking SSM, completely spike-based** without using any non-linearities such as GeLU, GLU, or GSU.
 - Furthermore, we observed that SHaRe-SSM **outperforms all first-order SSMs on very long sequence classification** and is more energy efficient than ANN-based second-order SSMs.
 - Also, we extended SHaRe-SSM to energy-efficient **long-range regression** applications and proposed a **convolving kernel-based regression**. We observed that SHaRe-SSM outperforms first-order SSMs on a **50K sequence length regression task**.
 - Finally, we studied and explored the **impact of discretization methods** and network **heterogeneities** on SHaRe-SSM.

Publication: NeurIPS 2025 TS4H (Spotlight) - A Second-Order SpikingSSM for Wearables
Kartikay Agrawal, Abhijeet Vikram, Vedant Sharma, Vaishnavi N, Ayon Borthakur

We Proposed SHaRe-SSM for Wearable Devices giving SOTA performance on **Human Activity Recognition datasets and PPG-DalAi EEG datasets** while being extremely **energy efficient (52.1x)**.

Publication: ICML 2026 (Regular) - A Spiking Heterogeneous Harmonic Resonate-and-Fire State Space Model for Time Series - [Webpage](#), [Code](#)
Kartikay Agrawal, Vaishnavi N, Abhijeet Vikram, Vedant Sharma, Ayon Borthakur

- **Semester Project (Jan-Apr 2025): FER2013 - ResNet (Chaitanya Guttikar - IISER Pune)**
 Worked on **FER2013 dataset** with **ResNET like architecture** and achieved an accuracy of **71.25%** by correcting the class imbalance and cleaning the dataset - [Demo](#)
- **Winter Project (Dec-Jan 2025): Neuromorphic Computing (Prof. Ayon Borthakur - IIT Guwahati)**
 Studied **Spiking Neural Networks** and its comparison with Artificial Neural Networks and the **changes and benefits of Learnable Initial Membrane Potential and other Heterogeneities** - [Link](#)
- **Summer Project (Jun-Jul 2025): Evolutionary Biology (Prof. Sutirth Dey - IISER Pune)**
 Standardized the Spatial Kernel with Drosophila Activity Monitor (DAM) and its activity data analysis. Also built an **Drosophila larvae activity tracker** to study the evolutionary selection pressure of adult dispersal on larval dispersal ability.
Key Takeaways - Data Cleaning and Data analysis using Scipy, Numpy, matplotlib, OpenCV
- Winter Project (Dec-Jan 2023) Evolutionary Biology (Prof. Sutirth Dey)
 Standardized few techniques (DIET, CAFE) to measure dietary intake of outbred Drosophila
- Other Research Projects -
 - Effects of light wavelengths as insect attractants for pests and pollinators.
 - **Mouse training and Mouse Brain Surgery** for brain calcium imaging.
 - **PCR** to amplify TERT gene (linked to ageing and oncogenesis) with SNP and extract it.
 - Effect of Climate change on **Butterfly dispersal** - **Cox's Proportionality Hazard Model**.
 - Studied **fly circadian rhythm** changes after crossing GMR GAL – 4 × UAS lines and their data analysis.
 - Review on **Earthquake location determination methods** - Non-Linear Methods

Skills & Interests

- **Programming: Python** (scipy, Matplotlib, Numpy, Pandas, **torch**, **Jax**), Git, Latex
- **Machine Learning:** Supervised & Unsupervised Learning, Model Evaluation, Feature Engineering, benchmarking, optimisation
- **Deep Learning:** CNN, ResNet, VGGNet, UNet, RNN, LSTM, **State Space Models (SSMs)**, **Diffusion Models**.
- **Worked with :** Neuromorphic Computing, Spiking Neural Networks (SNNs), Low power AI systems, Mathematical Finance, PINNs, NLD and Random Matrix Theory

Acomplishments

- **NEET (2022)**- Gen AIR 30156
- **IISER Aptitude Test (IAT)** - Gen AIR 534
- **Ramanujan Memorial Mathematics Contest** (17-18, 18-19) - CHS (BHU) Zonal Topper
- **PRMO** (Homi Bhabha Centre for Science Education) (qualified for RMO), Regional Mathematics Olympiad (2019)
- **National Standard Examination in Junior Sciences** (HBCSE)

Hobbies and Achievements

- **Basketball**
CBSE Silver Medal - Basketball under 17 Cluster
- **Hindustani Classical - Flute**
Senior Diploma in Hindustani Classical Music Prayag Sangeet Samiti, Prayagraj, with distinction in theory

